Quasi-Static Elastography-Driven Automated Robotic Ultrasound Screening and Localization

Hanying Liang, Shipeng Zhang, Guochen Ning, *Member, IEEE*, Jianwen Luo, *Senior Member, IEEE*, and Hongen Liao, *Senior Member, IEEE*

***Abstract*—Robotic ultrasound imaging systems primarily focus on enhancing automation in grayscale image acquisition but lack essential functional information, which restricts their clinical effectiveness and efficiency. In this regard, we propose a novel robotic ultrasound system capable of automatically screening and anomaly localization using quasi-static elastography (QE). For continuous screening, a compliant force control strategy is devised to manage the complex probe operations required for elasticity data acquisition. This involves the integration of adaptive out-of-plane posture control and in-plane palpation motion control. For anomaly localization, tissue strains are analyzed using multi-source motion data. We introduce an unsupervised tissue displacement estimation method, complemented by a strain estimator with multi-frame fusion for robust strain estimation. A 3D strain map is reconstructed to enable closed-loop control for automated robotic localization. The system has been validated through extensive experiments on two realistic phantoms and tested on human subjects. Results demonstrate that our system can perform robotic QE-based screening across subjects with varying surface conditions and lesion depths, showing improved efficiency and adaptability compared to existing systems. It achieves satisfactory accuracy in strain-based anomaly localization, with a detection rate of 0.77 and an average localization error of 1.01 ± 0.47 mm for the abdominal phantom, and 0.73 and 3.44 ± 0.84 mm for the more challenging thyroid phantom. By identifying and localizing suspicious anomalies in 3D space, the proposed system shows promise in providing preliminary diagnostic assistance in superficial tissue ultrasound examinations.**

***Index Terms*—Complex medical operation, robotic ultrasound system, quasi-static elastography, feedback control, strain estimation.**

I. Introduction

AUTOMATIC robotic ultrasound imaging systems (RUISs) have attracted growing research interest in recent years, offering distinct benefits in terms of alleviating sonographer workload, reducing operator dependency, and delivering consistent imaging results. While grayscale images remain the primary modality in current clinical practice and RUISs, several advanced functional modalities, such as Doppler ultrasound and elastography, have been explored to enhance ultrasound's diagnostic capabilities [1]. However, integrating automated robotic control with advanced ultrasound imaging techniques requires not only the fundamental functionality to ensure proper acoustic coupling but also specific probe motion to acquire additional information, such as a tilted probe posture for detecting Doppler signals [2]. Therefore, it highlights the need for further research to integrate functional imaging modalities into the development of robotic ultrasound systems.

Among these advanced modalities, elastography stands out as a non-invasive and versatile approach that assesses pathology by measuring tissue elasticity. It serves as a valuable complement to grayscale ultrasound, improving its sensitivity and specificity, particularly in the diagnosis of superficial soft tissues [3], and thereby reducing unnecessary biopsies [4]. The basic principle of elastography involves measuring the response of tissues to internal or external excitation, which can be broadly categorized into dynamic and quasi-static. Dynamic elastography allows for quantitative measurements by directly assessing the propagation speed of shear waves. However, it requires specialized hardware to generate shear waves using acoustic radiation force (ARF) [5] or external vibrators [6], [7]. Effective dynamic elastography is limited to a small region of interest (ROI), due to shear wave signal attenuation over distance and the substantial computational demands of RF data processing. Significant variations in elasticity measurements arise from ROI selection, which remains a critical issue in this field [8]. Quasi-static elastography (QE) measures tissue elasticity by analyzing the strain distribution within tissues under an externally applied quasi-static force. Owing to its straightforward principles and minimal device requirements, QE is capable of deeper, frame-sized strain estimation and can be implemented on any commercial ultrasound

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Hanying Liang, Shipeng Zhang, Jianwen Luo, and Hongen Liao are with the School of Biomedical Engineering, Tsinghua University, China.

Guochen Ning is with the School of Clinical Medicine, Tsinghua University, Beijing 10084, China (e-mail: ningguochen@tsinghua.edu.cn).

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device. In superficial tissue diagnosis, QE has demonstrated performance comparable to shear wave-based elastography (SWE) in some clinical studies [9]. Therefore, QE offers a balanced solution with advantages in accessibility, cost, and practical use, and is used in this work.

However, continuous QE scanning requires complex probe operation, involving alternating between an initial adjustment to achieve perpendicular contact and a subsequent constrained palpation motion along the axial direction. This process is challenging on soft surfaces with inherent uncertainty and complexity. In clinical practice, such operations are performed manually. Data decorrelation due to out-of-plane motion during palpation is the major problem in freehand QE, leading to image artifacts and degraded strain quality. Moreover, the demanding control imposes an operational burden on sonographers, often limiting freehand QE to a two-dimensional (2D) plane and leading to unrepeatability [10].

To overcome the above limitations, robotic elastography presents a promising solution due to its stable control and advanced sensing capabilities. Early efforts]integrated QE with da Vinci surgical robots for minimally invasive surgery [11]. Napoli et al. [12] introduced a robot-assisted probabilistic elasticity mapping framework using an extended Kalman filter, but it requires 20 seconds (s) to obtain a pair of compressed data. The robot primarily assisted in-plane palpation for 2D QE, while other degrees of freedom (DoF) of the probe motion were not automated. Besides, the validation was limited to a simple cubic gelatin phantom. Patlan-Rosales et al. [13] developed systems for automatic palpation and stiff-target centering using QE feedback with a 6-DoF robot. But manual intervention was still required to select the ROI and adjust probe orientation via teleoperation. It took around 2 minutes for robotic centering of the lesion target. In these setups, the probe was manually placed over the abnormal area, assuming perpendicular contact and a precisely known lesion location. Such prior information is often unavailable before examinations. Automating palpation motion alone is insufficient for wide-area QE-based screening and may compromise strain quality. Clinical validation insights highlight that enhancing the automation of probe control subtasks, providing efficient diagnostic assistance, and ensuring technique accessibility are key directions for advancing robotic ultrasound into clinical applications [14].

To automate probe control and ensure perpendicular contact for QE, Groenhuis et al. [15] used magnetic resonance images to determine optimal probe orientations for axial palpation on a breast phantom. The elastograms were only reconstructed at a limited number of discrete locations, rather than through continuously scanning along the complex surface. Chen et al. [16] developed a robot-assisted wireless strain imaging system that combined palpation with stepwise scanning. Yao et al. [17] leveraged a depth camera to facilitate 3D QE based on the surface model. The systems were evaluated only by moving the probe linearly over a flat surface. The simple position control law used in these systems lacks the flexibility to adapt to complex soft surfaces. Additionally, the external vision-based control methods inevitably suffer from optical occlusion and surface deformation. In contrast, the proposed method relies solely on force information to achieve compliant probe control, enabling efficient robotic screening with adaptability across varying subjects, as validated through extensive experiments.

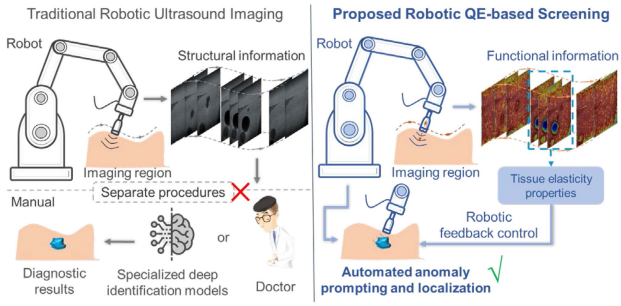


Fig. 1. Difference between the paradigms of RUIS and the proposed system. Existing RUISs treat robotic imaging and diagnostic interpretation as separate procedures, while our system integrates them into a single robotic control framework by incorporating functional information reflecting tissue elasticity.

Recent advancements in RUISs have primarily focused on automating grayscale image acquisition. A typical framework of RUISs is illustrated Fig. 1. Although various information have been integrated to ensure desired imaging content and quality, image acquisition and their diagnostic interpretation are still treated as separate processes, rendering it inefficient for practical applications. While certain RUISs have been tailored for specific organ [18], these anatomical structures alone do not provide direct diagnostic insight from a clinical perspective. The disconnect between probe motion and data analysis results in grayscale images with limited information, making differentiation between normal and pathological tissues challenging. Consequently, diagnostic interpretation remains ambiguous post-scanning, typically requiring intervention by experienced sonographers or specialized deep learning models for lesion identification. Imaging techniques such as QE that directly reveal tissue properties can improve efficiency and objectivity. This additional information holds significant clinical value but is often overlooked in current RUISs.

Recent learning-based elastography has shown that accurate displacement/strain can be obtained without ground-truth labels by leveraging image similarity and physics-inspired priors. Delaunay et al. enforced spatio-temporal consistency to stabilize quasi-static strain estimation [19]. Wen et al. proposed a CNN-based speckle-tracking pipeline trained purely with unsupervised losses to recover dense motion from B-mode/envelope sequences [20]. Tehrani et al. explored semi-/unsupervised schemes tailored to USE, introducing physically inspired constraints (e.g., smoothness, Poisson-ratio coupling) that notably improve lateral motion and strain quality [21], [22]. More recently, RAFT-USENet adapts the RAFT architecture to the high-frequency ultrasound domain, unifying accurate axial and lateral displacement estimation under physics-informed regularization [23]. While the above methods are optimized for small, near-linear inter-frame motions—typically with RF inputs—our setting features larger, non-uniform deformations and B-mode-only access, utilizing a transformer model to capture long-range context and a multi-frame fusion strategy. We bridge this gap by coupling compliant, automatically controlled

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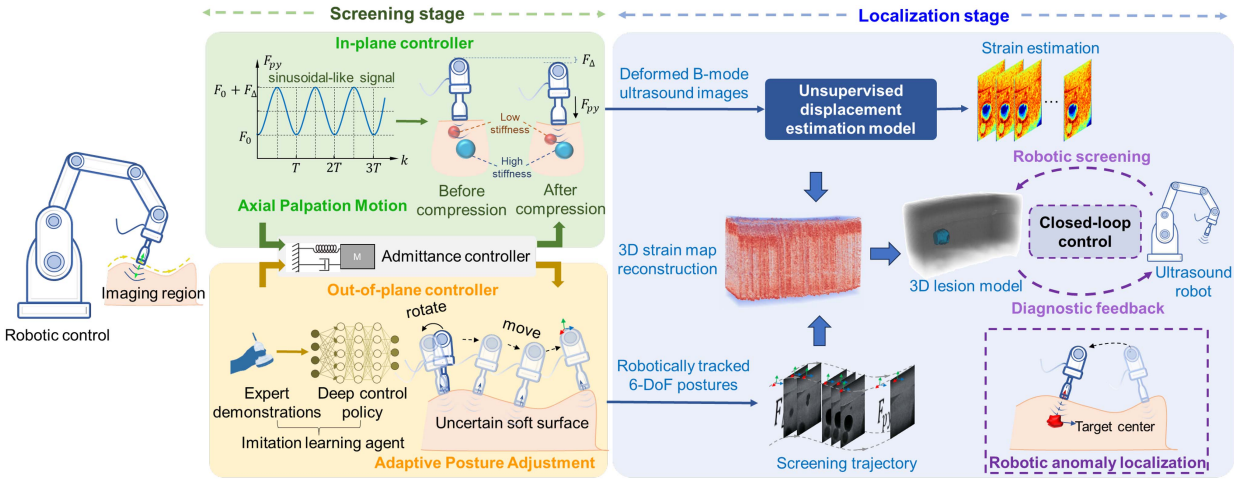


Fig. 2. The overall proposed framework. In the screening stage, the compounded probe motion is controlled by an out-of-plane controller to ensure perpendicular contact and an in-plane controller to perform axial palpation. In the localization stage, tissue strains are estimated based on outputs of an unsupervised displacement estimation model. 3D tissue elasticity information is used for closed-loop control to guide the robotic anomaly localization process.

palpation with unsupervised displacement and strain estimation within a robotic screening and localization framework.

In this work, we present a novel robotic ultrasound framework that enables automated screening and anomaly localization using QE. Its distinction from existing RUISs is illustrated in Fig. 1. By assessing tissue elasticity and integrating it into a feedback control loop, the proposed framework unifies automated ultrasound image acquisition and diagnostic interpretation within a robotic system. The main contributions of this work are summarized as follows:

1) We introduce a novel paradigm in robotic ultrasound that utilizes QE to obtain functional information. The system enables wide-area QE-based screening and provides direct prompts indicating the presence and location of anomalies.
2) We couple robotic probe control with data analysis to obtain tissue elasticity. A compliant force control strategy is presented to automate the complex probe operations required for effective QE, integrating both out-of-plane and in-plane control with enhanced adaptability and efficiency.
3) We present a robust and unsupervised pipeline for strain estimation through multi-source data analysis. Operating on broadly accessible B-mode streams, our method is compatible with commercially available ultrasound devices, maintaining hardware simplicity in standard configurations.
4) Extensive experiments were conducted on both realistic phantoms and human subjects to validate the system's feasibility. The results demonstrate its applicability across diverse subjects with uncertain soft surfaces, highlighting its potential for superficial tissue ultrasound examinations.

II. System and Control Framework

Our system mimics the clinical freehand QE examination process while enhancing it with automated robotic control. As shown in Fig. 2, the workflow is divided into the screening stage and the localization stage. Before the examination, the presence and location of anomalies are unknown. Hence, a screening process is first performed to broadly survey the region using QE to detect potential abnormalities with lower elasticity. To automate the complex imaging operation, we develop a compliant force control strategy that integrates both in-plane and out-of-plane controllers. The in-plane controller is tasked with axial palpation, using variable force control to generate sinusoidal pressure that induces tissue displacement indicative of elasticity. The out-of-plane controller adaptively adjusts the probe posture to ensure perpendicular contact with the soft surface. The coordinated motion of both controllers enables efficient QE screening over a wide area. Multi-source data, including ultrasound images, corresponding probe poses, and contact forces are recorded for subsequent analysis.

In the localization stage, the probe is repositioned over the suspicious area upon identification. Tissue strains are estimated to offer intuitive insights for anomaly identification. We introduce an unsupervised method to predict displacement fields between deformed B-mode ultrasound images. A strain estimator using multi-source and multi-frame data is proposed for robust strain estimation, transforming tactile information into intuitive visual cues. Regions of low elasticity are identified and extracted for robotic feedback control, enabling automatic anomaly localization. In this way, the proposed framework establishes a closed loop between robotic ultrasound imaging and diagnostic interpretation.

A. Out-of-Plane Controller for Adaptive Posture Adjustment

To ensure effective QE, it is crucial that the acquired ultrasound images consistently remain within the initial axial-lateral plane and exhibit only axial displacements. This requires maintaining perpendicular contact and proper acoustic coupling between the probe and soft surfaces. To avoid explicit modeling of uncertain imaging environments, the out-of-plane controller is

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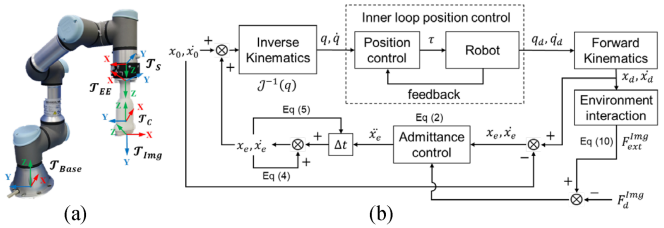


Fig. 3. Associated Cartesian frames (a) and the proposed force scheme (b).

built using an imitation learning-based policy [24], inheriting the adaptability of human demonstrations. It relies solely on contact force information, mitigating instability caused by optical occlusions and interference encountered with vision-based methods. Additionally, it preserves hardware simplicity by eliminating the need for external devices.

The policy is first trained using reinforcement learning (RL), with a predefined reward function designed to minimize probe torque for satisfying the perpendicular contact condition. The state $s_t = \{\boldsymbol{\mathcal{F}}_{t-k+1}, \boldsymbol{\mathcal{F}}_{t-k+2}, .., \boldsymbol{\mathcal{F}}_t\}$ consists of k continuous observations of the sensed wrench $\boldsymbol{\mathcal{F}}_t = [F, \boldsymbol{\Gamma}] \in \mathbb{R}^6$ to enhance contextual information, where $\boldsymbol{F} \in \mathbb{R}^3$ denotes the applied force and $\boldsymbol{\Gamma} \in \mathbb{R}^3$ denotes the corresponding torque. The policy predicts the action $a_{RL} = \{(\boldsymbol{\mathcal{F}}_{ext}^{RL})\}$, which determines the probe's 3-DoF posture by applying the desired contact wrench $\boldsymbol{\mathcal{F}}_{ext}^{RL} = [0, F_y, 0, \Gamma_x, 0, \Gamma_z]$.

Given that the manually defined objective is prone to suboptimality and lacks generalizability, the policy is further optimized to generate likelihood trajectories by learning from a set of demonstrations $\mathcal{D} = \{\tau_1, \tau_2, \ldots, \tau_N\}$, where $\tau = \{(s_1, a_1), \ldots, (s_T, a_T)\}$ represents a complete operational trajectory. Human demonstrations, obtained by manually maneuvering the probe to refine the perpendicular posture predicted by the RL policy, are incorporated to enhance policy transferability and stability [24]. The optimization is formulated as an adversarial process:

$$\nabla_\varepsilon J(\varepsilon) \approx \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\mathcal{D}} \left[\nabla_\varepsilon r_\varepsilon (s_t, a_t, s_{t+1})\right]}_{Human\ demonstration} - \underbrace{\log \pi_\varepsilon (a_t | s_t)}_{RL\ policy} \tag{1}$$

A 6-DoF force-velocity action space, defined as $a_{IL} = \{(\boldsymbol{\mathcal{F}}_{ext}^{IL}, \boldsymbol{\mathcal{v}}_{ext}^{IL})\} = \{(F_y, \Gamma_{r_x}, \Gamma_{r_z}, v_y, w_{r_x}, w_{r_z})\}$, is established to integrate rich control information from demonstrations. In the absence of large surface motion, the out-of-plane controller adaptively adjusts probe posture using the learned wrench and velocities, ensuring stable and perpendicular contact on uncertain soft surfaces for subsequent axial palpation.

B. In-Plane Controller for Axial Palpation Motion

The controllers are based on admittance control to perform compliant probe motion actions. The associated Cartesian frames are shown in Fig. 3(a). In-plane motion, in the context of ultrasound imaging, corresponds to the x, y plane, i.e., the imaging plane of the probe. Following QE principles, an external compressive force is applied along the initial axial direction (y axis) in the image frame $\mathcal{T}_{Img}$ to obtain pre- and post-compressed image pairs. Taking the position part as an example, admittance control aims to achieve the following dynamics in the image frame $\boldsymbol{\mathcal{T}}_{Img}$:

$$\boldsymbol{M}\left(\ddot{x_d} - \ddot{x_0}\right) + \boldsymbol{B}\left(\dot{x_d} - \dot{x_0}\right) + \boldsymbol{K}\ (x_d - x_0) = \boldsymbol{F}_{ext}^{Img} \tag{2}$$

where $\boldsymbol{M}$, $\boldsymbol{B}, \boldsymbol{K}$ denote respectively the inertial, damping, and the stiffness matrix, $\boldsymbol{F}_{ext}^{Img}$ is the external force, x_0, x_d represent the initial and desired position of the probe. $\dot{x_d}, \ddot{x_d}$ and $\dot{x_0}, \ddot{x_0}$ are the first and second-order derivatives of x_d and x_0, respectively. The controller regulates the probe movements by calculating the deviations and adjusting to track the desired motion parameters based on the external force:

$$\ddot{x_e}^{\,t} = \boldsymbol{M}^{-1}\left(\boldsymbol{F}_{ext}^{Img} - \boldsymbol{B}\dot{x_e}^{\,t} - \boldsymbol{K}x_e{}^{t}\right) \tag{3}$$

$$\dot{x_e}^{\,t+1} = \dot{x_e}^{\,t} + \ddot{x_e}^{\,t}\Delta t \tag{4}$$

$$x_e{}^{t+1} = x_e{}^{t} + \dot{x_e}^{\,t+1}\Delta t \tag{5}$$

where $x_e = x_d - x_0$ denotes the position deviation, and Δt is the control cycle. The orientation deviation is computed using $r_e = r_d\ r_0{}^T$. A force sensor is affixed to the robot end-effector to measure the 6-DoF contact force $\boldsymbol{\mathcal{F}}_m^S$ within the sensor frame $\boldsymbol{\mathcal{T}}_S$. Gravity compensation and force transformation are performed to obtain the actual contact force. The external wrench $\boldsymbol{\mathcal{F}}_{ext}^{Img}$ can be represented as:

$$\boldsymbol{\mathcal{F}}_{ext}^{Img} = \boldsymbol{\mathcal{T}}_{Img}^{S}{}^{T}\left(\boldsymbol{\mathcal{F}}_m^S - \boldsymbol{\mathcal{T}}_C^S \boldsymbol{\mathcal{G}}^C\right) \tag{6}$$

where $\boldsymbol{\mathcal{G}}^C = [0, 0, -m\mathcal{g}, 0, 0, 0]^T$ is the gravity force vector expressed in frame $\boldsymbol{\mathcal{T}}_C$, which is aligned with the center of mass of the overload and shares the same orientation as the robot base frame $\boldsymbol{\mathcal{T}}_{Base}$. $\mathcal{g}$ denotes the acceleration of gravity, and m denotes the mass of the load, which includes the probe and the sensor. $\boldsymbol{\mathcal{T}}_A^B \in \mathbb{R}^{6\times6}$ denotes the force twist transformation matrix from frame $\boldsymbol{\mathcal{T}}_A$ to frame $\boldsymbol{\mathcal{T}}_B$. The transformation relationship can be derived from:

$$\boldsymbol{\mathcal{T}}_{Img}^S = \boldsymbol{\mathcal{T}}_{Base}^S\, \boldsymbol{\mathcal{T}}_{Base}^{EE}{}^{-1} \boldsymbol{\mathcal{T}}_{Img}^{EE} \tag{7}$$

$$\boldsymbol{\mathcal{T}}_C^S = \boldsymbol{\mathcal{T}}_{Base}^S\, \boldsymbol{\mathcal{T}}_{Base}^{C}{}^{-1} \tag{8}$$

where $\boldsymbol{\mathcal{T}}_{Base}^S, \boldsymbol{\mathcal{T}}_{Base}^C, \boldsymbol{\mathcal{T}}_{Base}^{EE}$ can be obtained via load detection and robot forward kinematics. $\boldsymbol{\mathcal{T}}_{Img}^{EE}$ is the transformation between the end-effector frame and the image frame, determined via point-based robotic ultrasound calibration. To facilitate consistent and continuous axial palpation at different positions, the in-plane controller generates a sinusoidal-like force signal along the y-axis in the image frame $\boldsymbol{\mathcal{T}}_{Img}$:

$$F_{py}\ (k) = F_0\ + F_\Delta \sin\left(\frac{2k\pi}{T_p} + \frac{\pi}{2}\right) \tag{9}$$

where F_0 is the offset value, F_Δ is the amplitude of the signal variation, and T_p, k are the period and the number of samples of the discrete signal, respectively. The overall force control scheme is shown in Fig. 3(b). Let F_{py} be the y-axis component of

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the desired contact force $\boldsymbol{F}_d^{Img} = \boldsymbol{F}_p^{Img}$ to achieve the intended palpation motion:

$$\ddot{x_e} = \boldsymbol{M}^{-1}\left[\left(\boldsymbol{F}_{ext}^{Img} - \boldsymbol{L_p} * \boldsymbol{F}_p^{Img}\right) - \boldsymbol{B}\,\dot{x_e} - (\boldsymbol{I}_3 - \boldsymbol{L_p})\,\boldsymbol{K}x_e\right] \tag{10}$$

where $*$ denotes element-wise multiplication, $\boldsymbol{I}_3 \in \mathbb{R}^3$ is the identity matrix, and $\boldsymbol{L_p} \in \{0,1\}$ is a selection vector. We configure translation along the y-axis as the only compliant motion by setting $\boldsymbol{L_p} = [0,1,0,0,0,0]^T$, while maintaining impedance control in other DoF. This constrains the probe motion to the surface's perpendicular direction while suppressing non-axial movements during palpation, thereby enhancing the quality of strain estimation.

C. Force Control Strategy With Action Fusion

We fuse the actions of the controllers to efficiently execute the probe operation required for QE-based screening. At the beginning, the probe makes initial contact with the surface in a random posture, often significantly misaligned with the desired posture. Hence, the out-of-plane controller initiates a multi-step adjustment process to ensure the probe achieves the perpendicular posture at the very first contact. This process terminates either upon reaching a preset maximum number of adjustments N_a or upon meeting the specified condition:

$$\Omega\ (\boldsymbol{\Gamma}) = \begin{cases} \text{True}, & |\Gamma_x| \le \rho_\gamma,\ |\Gamma_y| \le \rho_\gamma \\ \text{False}, & \text{others} \end{cases} \tag{11}$$

where Ω serves as the termination indicator of the multi-step adjustment process. When the probe reaches a perpendicular posture, the torques around both the x and y axes should be zero. Thus, the threshold ρ_γ is empirically set to a small value of 0.01. We consider $\Omega = True$ when the probe reaches the desired posture necessary for in-plane palpation.

The force control strategy, enabling seamless transitions between out-of-plane and in-plane motion control, is illustrated in Algorithm 1. Although multi-step adjustments result in a more accurate perpendicular posture, applying them at every waypoint can be time-consuming. Considering the continuous nature of the imaging surface, only a single-step adjustment is performed at each waypoint during screening. Through action fusion and the shared force control scheme shown in Fig. 3(b), the proposed strategy ensures continuous QE-based robotic screening on uncertain soft surfaces with improved efficiency.

Algorithm 1: Force Control Strategy With Action Fusion.

Input: planned imaging trajectory with J waypoints, $\boldsymbol{\mathcal{F}}_p^{Img}$, N_a, n=0
Initialize: Make contact with the surface in a random posture
  $\boldsymbol{\mathcal{F}}_0 = \boldsymbol{F}_0, \boldsymbol{\Gamma}_0, \boldsymbol{F}_0 \in \mathbb{R}^3, \boldsymbol{\Gamma}_0 \in \mathbb{R}^3, x_0, \dot{x_0}, \ddot{x_0}$
for each waypoint j= 1 . . . J **do**
  < ***Adaptive Posture Adjustment*** >
  do
    Data sensing: $\boldsymbol{\mathcal{F}}_t = \boldsymbol{F}_t, \boldsymbol{\Gamma}_t$
    Out-of-plane controller: $a_{IL} = \{(\boldsymbol{\mathcal{F}}_{ext}^{IL}, \boldsymbol{v}_{ext}^{IL})\}$, $\boldsymbol{L}_o = \{0,1,0,1,0,1\}$
    $\ddot{x_e}^t = \boldsymbol{M}^{-1}[(\boldsymbol{\mathcal{F}}_{ext}^{Img} - \boldsymbol{L}_o * \boldsymbol{\mathcal{F}}_{ext}^{IL}) - \boldsymbol{B}\dot{x_e}^t - (\boldsymbol{I}_3 - \boldsymbol{L}_o)\boldsymbol{K}x_e{}^t]$
    $\dot{x_e}^{t+1} = \dot{x_e}^t + \dot{x}_{IL}^t + \ddot{x_e}^t \Delta t$
    $x_e{}^{t+1} = x_e{}^t + \dot{x_e}^{t+1} \Delta t$
  until $j > 1$or $(n > N_a$ or $\Omega(\boldsymbol{\Gamma}_t))$
  <***Axial Palpation Motion***>
  for each time $k = 1 \ldots \mathrm{T}_p$ **do**
    Data sensing: $\boldsymbol{\mathcal{F}}_t = \boldsymbol{F}_t, \boldsymbol{\Gamma}_t$
    In-plane controller: $\boldsymbol{\mathcal{F}}_p^{Img}(k)$, $\boldsymbol{L}_p = \{0,1,0,0,0,0\}$
    $\ddot{x_e}^t = \boldsymbol{M}^{-1}[(\boldsymbol{\mathcal{F}}_{ext}^{Img} - \boldsymbol{L}_p * \boldsymbol{\mathcal{F}}_p^{Img}) - \boldsymbol{B}\dot{x_e}^t - (\boldsymbol{I}_3 - \boldsymbol{L}_p)\boldsymbol{K}x_e{}^t]$
    $\dot{x_e}^{t+1} = \dot{x_e}^t + \ddot{x_e}^t \Delta t$
    $x_e{}^{t+1} = x_e{}^t + \dot{x_e}^{t+1} \Delta t$
  end for
  end for

III. Tissue Elasticity Information Acquisition

A. Unsupervised Tissue Displacement Estimation in 2-D B-Model Ultrasound Images

In strain imaging, displacement fields are typically derived through cross-correlation tracking of radio-frequency (RF) data before and after probe compression [8]. However, extensive RF data requires significant computational resources, limiting their use in real-time applications and within selected ROIs [11]. Working on local windows hinders the accurate prediction of large displacements and reduces robustness against global decorrelation due to out-of-plane motion [19]. Estimating large displacements benefits strain imaging by increasing strain contrast through larger external compressions [25]. RF data generally yield higher displacement accuracy for small deformations but are more vulnerable to decorrelation as deformations grow; in contrast, envelope/B-mode representations, after demodulation, offer a wider dynamic range and are thus preferable for large-deformation tracking. [26], [27]. RF data is not always accessible on commercial devices. Envelope data, with less high-frequency oscillations, are more robust to signal decorrelation at high strains and allow for frame-sized strain estimation [26].

Therefore, we employ the standard B-mode stream to estimate tissue displacement. This task presents inherent challenges, including speckle artifacts, noise, and low image contrast in ultrasound images. Densely annotating tissue displacements is laborious and impractical. To address these issues, we introduce an unsupervised ultrasound image displacement estimation method based on the transformer architecture [28], as depicted in Fig. 4. The moving and fixed ultrasound images, I^m and I^f, are first tokenized and fed into a transformer-based encoder. Leveraging the extracted hierarchical deep features, a convolutional decoder progressively predicts the displacement field ϕ between the deformed B-mode images. Attention mechanisms

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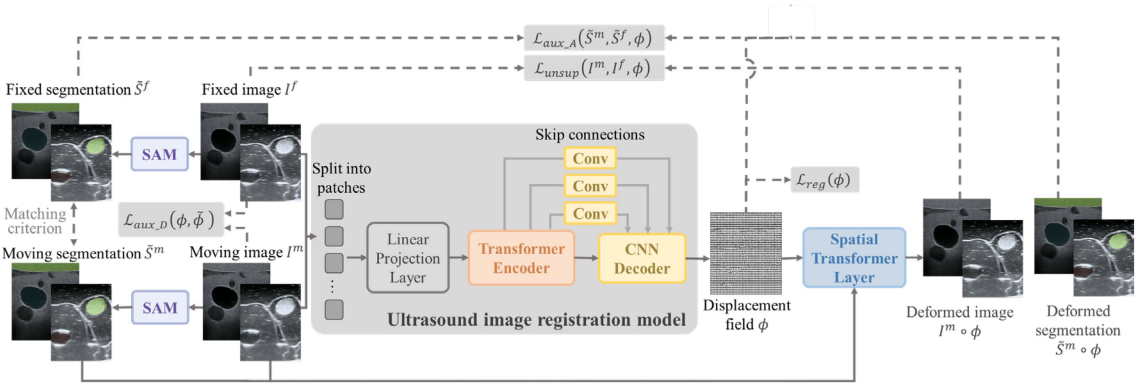


Fig. 4. The proposed unsupervised ultrasound image displacement estimation method. The model takes deformed B-mode ultrasound image pairs as input and predicts a dense displacement field. During training, auxiliary information on tissue displacement and anatomical semantics is integrated to guide the training process. Masks are shown only to illustrate the training-time auxiliary; no SAM segmentation is required at inference.

and long-range modeling enhances its ability to capture large and non-linear deformations.

General Unsupervised Displacement estimation: Unsupervised displacement estimation aims to align the fixed and moving images without relying on labeled data. The overall loss function consists of three parts: a classical unsupervised learning term $\mathcal{L}_{unsup}$, a regularization term $\mathcal{L}_{re}$, and a self-supervised auxiliary term:

$$\mathcal{L} = \mathcal{L}_{unsup} + \alpha \mathcal{L}_{reg} + \beta \mathcal{L}_{aux} \tag{12}$$

where α and β are weighting parameters. Given the inherent complexity of ultrasound images, we use the structural similarity index measure (SSIM) [29] to penalize the structural difference between the warped moving and the fixed images:

$$\mathcal{L}_{unsup}\left(I^m, I^f, \phi\right) = 1 - SSIM\left(I^m \circ \phi, I^f\right) \tag{13}$$

where $\circ$ is the warping operation that deforms the moving image I^m using a spatial transformer layer. Optimizing the unsupervised term alone can achieve high displacement estimation accuracy but may also lead to unrealistic predictions. Hence, we apply bending energy [30] to enforce smoothness and continuity:

$$\begin{aligned}\mathcal{L}_{reg}(\phi) &= \sum_{\boldsymbol{p}\epsilon\Omega} \left\|\nabla^2 \phi(\boldsymbol{p})\right\|^2 \\ &= \sum_{\boldsymbol{p}\epsilon\Omega} \left[\left(\frac{\partial^2 \phi(p)}{\partial x^2}\right)^2 + \left(\frac{\partial^2 \phi(p)}{\partial y^2}\right)^2 \right. \\ &\quad \left. + 2\left(\frac{\partial^2 \phi(p)}{\partial x \partial y}\right)^2\right]\end{aligned} \tag{14}$$

where the spatial gradients $\nabla\phi(p)$ are approximated using the forward differences between neighboring pixels $\frac{\partial \phi(\boldsymbol{p})}{\partial\{x,y,z\}} \approx \phi(p_{[x,y,z]}+1) - \phi(p_{[x,y,z]})$. The second-order regularizer not only penalizes sharply curving displacements but also constrains smoothness in the generated strain map.

Auxiliary Information Guidance: While the Transformer network excels in modeling complex displacements, its effectiveness relies on a large volume of training data. To improve data efficiency, we introduce auxiliary information to facilitate model training in a self-supervised manner. Pseudo labeling stands out as a popular technique to warm up the training process and boost model performance [31]. In this context, we introduce two types of pseudo labels incorporating both displacement and anatomical information:

$$\begin{aligned}\mathcal{L}_{aux} &= \mathcal{L}_{aux-D}\left(\phi, \tilde{\phi}\right) + \mathcal{L}_{aux-A}\left(\tilde{S}^m, \tilde{S}^f, \phi\right) \\ &= \omega_1 \left\|\phi - \tilde{\phi}\right\|_1 + \omega_2 \sum_i \mathcal{D}ice\left(\tilde{S}^m_{\Theta(i)} \circ \phi, \tilde{S}^f_i\right)\end{aligned} \tag{15}$$

where ω_1 and ω_2 are weighting parameters, and $\tilde{S}^m$, $\tilde{S}^f$ are the segmentation masks of I^m, I^f. The first term computes the L1 loss between the pseudo label and the predicted displacement field. Due to posture control errors, the probe may misalign perpendicular to the imaging surface, leading to non-axial movements during palpation. This introduces out-of-plane displacements, which negatively affect strain estimation. To avoid training an additional teacher network, we use the Lucas-Kanade (LK) optical flow algorithm [32] to generate the pseudo label $\tilde{\phi}$ from the deformed B-mode pairs. This injects a local photometric consistency prior and serves as regularization, nudging the model towards coherent in-plane displacements under quasi-static compression.

The second term assists in anatomical semantic perception by aligning segmentation masks using the predicted displacement field. To avoid laborious manual labeling, we leverage the segmentation foundation model, SAM [33]. SAM is used only during training on phantom data as a weak auxiliary regularizer; the trained displacement network does not use SAM at inference and is directly applied to in-vivo data without any extra supervision or fine-tuning. Consequently, SAM cannot introduce inference-time failure modes. As SAM-generated segmentations of ultrasound images are often noisy and lack anatomical semantics, we propose a general matching criterion to establish valid correspondences:

$$\Theta(i) = \underset{j}{\text{argmax}}\ \text{DICE}\left(\tilde{S}^m_j \circ \phi, \tilde{S}^f_i\right) \tag{16}$$

where $\tilde{S}^i_f$ denotes the i-th segmentation mask of the fixed image, and $\tilde{S}^m_j$ denotes the j-th segmentation mask of the moving

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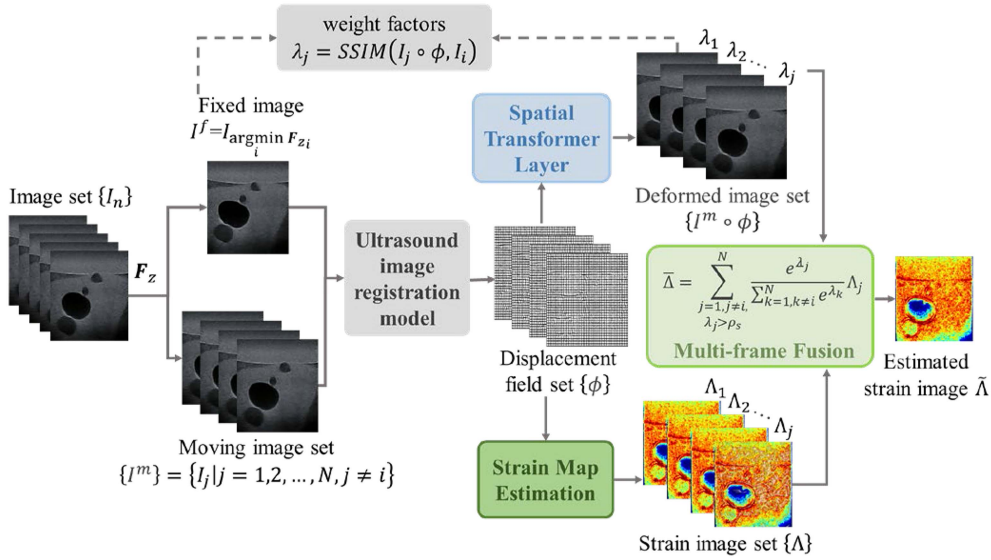


Fig. 5. Strain estimation with multi-frame fusion. Strain images are estimated and fused using sequential multi-source data recorded during axial palpation.

image. To improve robustness against noise and ambiguity, the segmentation masks are first filtered using a threshold ρ_a to retain only salient anatomies before matching. Consequently, it determines the match $\Theta(i)$ for $\tilde{S}_i^f$ by selecting the warped mask of the moving image with the highest Dice score to ensure consistent anatomical semantics.

B. Strain Estimator With Multi-Frame Fusion

Typically, the strain value is determined by calculating the spatial derivative of the directional displacement field $\mathcal{V}$ along the axial direction y:

$$\varepsilon_{yy} = \frac{\partial \mathcal{V}}{\partial y} \tag{17}$$

In addition to axial strain, previous research has investigated different strain indicators [29], such as lateral and shear strain components, to enrich the elasticity information. Specifically, shear strain exhibits greater robustness against decorrelated signals resulting from non-axial probe palpation motion [34]. Relying on individual components of the strain tensor can fall short of effectively distinguishing abnormalities. In this work, we adopt an equivalent (von-Mises–style) strain ε_{eq} that aggregates axial, lateral, and shear components:

$$\varepsilon_{eq} = \sqrt{{\varepsilon_{xx}}^2 + {\varepsilon_{yy}}^2 - \varepsilon_{xx}\varepsilon_{yy} + 3{\varepsilon_{xy}}^2} \tag{18}$$

$$\varepsilon_{xx} = \frac{\partial \boldsymbol{u}}{\partial x} \tag{19}$$

$$\varepsilon_{xy} = \varepsilon_{yx} = \frac{1}{2}\left(\frac{\partial \boldsymbol{u}}{\partial y} + \frac{\partial \boldsymbol{v}}{\partial x}\right) \tag{20}$$

where ε_{xx}, ε_{yy} are the normal strains along the x and y directions, ε_{xy}, ε_{yx}denote the shear strains, and $\mathcal{U}$, $\mathcal{V}$ denote the lateral and axial displacement, respectively. In the robotic palpation setting, small off-axis motions introduce non-axial displacements; ε_{eq} improves robustness to directional variability compared with axial-only strain. Noise from the lateral channel is mitigated by smoothness-regularized displacement estimation and SSIM-weighted multi-frame fusion. A least-square strain estimator [35] is applied to the displacements to diminish noise for improving strain quality.

We introduce a multi-frame fusion strategy to enhance robustness. The workflow is shown in Fig. 5. Deformed images $\{I_n\}$ under varying applied forces $\{\boldsymbol{F}_n\}$ are recorded during palpation. The image with the least axial force is designated as the fixed image $I^f = I_{\underset{i}{\text{argmin}}\, \boldsymbol{F}_{z\,i}}$, while the remaining images constitute the moving image set $\{I^m\} = \{I_j | j = 1, 2, \ldots, N, j \neq i\}$. The SSIMs between the fixed and the deformed moving images are measured to quantify displacement estimation accuracy, which serves as weight factors for multi-frame fusion:

$$\bar{\Lambda} = \sum_{\substack{j=1, j\neq i,\\ \lambda_j > \rho_s}}^{N} \frac{e^{\lambda_j}}{\sum_{k=1,k\neq i}^{N} e^{\lambda_k}} \Lambda_j \tag{21}$$

where $\lambda_j = SSIM(I_j \circ \phi, I_i)$, and Λ_j, $\bar{\Lambda}$ denote respectively the estimated strain images and fused results. The threshold ρ_s is defined to filter out strain images with low displacement estimation accuracy, as these may introduce artifacts. Compared to single-image methods, multi-frame fusion offers increased robustness against noise and displacement estimation inaccuracies.

C. Robotic Lesion Localization With Elasticity Feedback

The 3D strain map is reconstructed using robotically tracked 6-DoF postures. Areas exhibiting lower elasticity and strain are identified and extracted to build a 3D anomaly model for robotic feedback control. The probe is automatically repositioned to localize the anomaly and capture a centered image. Targeted at the model center, the probe reorients itself perpendicular to the surface using the out-of-plane controller while aligning with the

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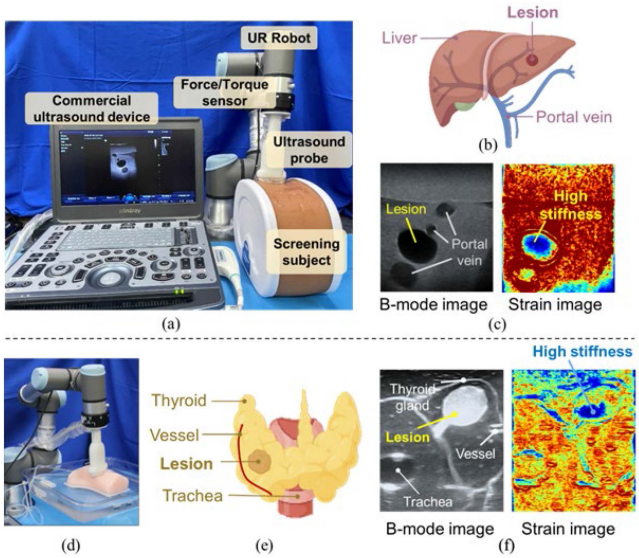


Fig. 6. System and experimental setups, including an abdominal phantom (a) and a thyroid phantom (d) with varying tissue elasticity, as shown in (b) and (e). While distinguishing tissue properties in traditional B-mode images, as shown in (c) and (f), is challenging and requires specialized knowledge, our method clearly identifies low-elasticity areas, shown in deep blue in the strain images.

intended imaging plane. The desired contact force is set to a constant value of 5 N. The adjustment process terminates when either the distances between the model center and the image centerline are below 1 mm or the cumulative adjustment steps reach the predefined maximum limit. Based on tissue elasticity information and feedback control, the system automatically localizes the anomaly, offering a centered image visualization for further evaluation.

IV. Experiments and Results

A. Experiment and System Setups

The system setup retains the basic configuration of general RUISs. It includes a lightweight robot (UR3, Universal robot, Denmark), a mounted linear probe (M8, Mindray, China), and a 6-DOF force sensor (FT300 Force/Torque Sensor, Robotiq, Canada). The system was validated on two realistic phantoms, as shown in Fig. 6(a) and (d), and qualitatively tested on human subjects. All experiments were approved by the local IRB (No. 20220152). Quantitative experiments were conducted on an abdominal phantom (Model 057A, CIRS Inc., USA) and a thyroid phantom (Model LYD822, LYD MED, China), representing deeper and superficial lesion cases, respectively. The imaging subjects contain multiple anatomical structures with varing elasticity, as shown in Fig. 6(b) and (e). Experimental settings for different screening subjects are listed in Table I. In traditional B-mode images, lesions may resemble normal tissues in appearance or size, making differentiation difficult. However, tissues with low elasticity, such as lesions and bone, can be intuitively identified in strain images estimated using our method, as shown in Fig. 6(c) and (f). A screening region was predefined for each subject. Threshold parameters ρ_a, ρ_d and ρ_s were empirically set to 100 pixels, 0.9, and 0.9, respectively. The robotic screening and localization process was repeated ten times on each phantom, with randomized trajectories and initial probe postures.

TABLE I
Imaging Subject and Parameter Configurations

Term	Abdominal phantom	Thyroid phantom	Human forearm
Target	Lesion	Lesion	Radius & Fascia Subcutaneous mass
Target depth [mm]	~ 40	~ 15	<20
Target size [Ø/mm]	10	10	/
Image depth [cm]	6	5	2
Image spacing [mm×mm]	0.76×0.76	0.63×0.63	0.25×0.25
Palpation force range	0~6	2~6	2~6

For training the displacement estimation model, a deformed B-mode image dataset was created. To mitigate the labor burden and errors from manual operation, we introduce a robot-assisted data collection method to automate this process. It involves a multi-step adjustment process followed by axial palpation to gather multi-source data across one palpation cycle. During data collection, the contact point remains static at each imaging position. We refer to this as the 'static' motion pattern, distinguishing it from the 'screening' process. Deformed images were collected at 120 distinct positions on each phantom, with data from 200 positions used for training and 40 for validation. The robot-assisted process took an average of 7.2 s per palpation cycle, collecting an average of 43 images per position. This demonstrates improved efficiency and stability compared to manual data collection. The acquired images were cropped and resized to 256×256. The displacement estimation model was trained for 400 epochs using the Adam optimizer, with a learning rate of 0.001 and a batch size of 16. The weighting parameters were empirically set to $\alpha = 0.1$, $\beta = 1$, $\omega_1 = \omega_2 = 0.5$ without hyperparameter tuning. All model computations were performed on a workstation (Intel Core i9-10980XE CPU, NVIDIA RTX 3060 GPU).

B. Evaluation of Probe Motion Control

We first analyzed the efficiency of the proposed force control strategy. The time distribution of a randomly selected robotic screening process on the two phantoms is showcased in Fig. 7(a), with the time evaluation across all trajectories listed in Table II. The out-of-plane controller requires more time for initial multi-step adjustments to achieve a perpendicular posture on the thyroid phantom, which exhibits greater curvature and softness. Single-step adjustments leverage the continuous nature of the surface and are not affected by changing surface conditions. Thus, the duration of the robotic screening process did not significantly increase, even on curved soft surfaces. The average time to complete a 10 cm robotic screening on the abdominal phantom is 117.66 ± 5.30 s, while a 7 cm screening on the thyroid phantom takes 76.18 ± 9.99 s. In-plane palpation takes slightly less time per cycle on the thyroid phantom due to the smaller amplitude of the sinusoidal-like force signal.

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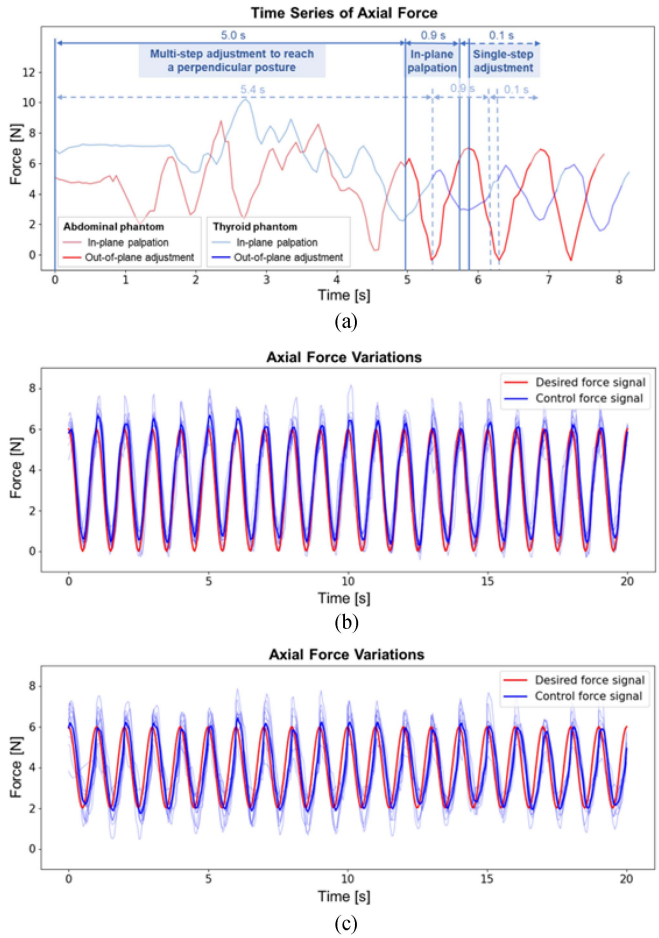


Fig. 7. Force control evaluation. (a) Force-time curves of a robotic screening process on the abdominal (red) and thyroid (blue) phantoms; Axial force variations during palpation on the abdominal (b) and thyroid (c) phantoms.

TABLE II
EVALUATION RESULTS OF THE PROPOSED FORCE CONTROL STRATEGY

Term	Evaluation metrics		Abdominal phantom	Thyroid phantom
Time consumption	multi-step adjustment [s]		5.18±0.71	5.71±0.51
	one-cycle palpation [s]		0.95±0.18	0.93±0.14
	single-step adjustment [s]		0.13±0.03	0.13±0.03
Force control accuracy	Static	ΔF_y [N]	0.70±0.48	0.92±0.76
		F_x [N]	0.19±0.51	0.79±0.61
		F_z [N]	0.21±0.36	0.33±0.42
	S i	ΔF_y [N]	0.74±0.55	1.05±0.65
		F [N]	0 62±0 84	1 58±0 79

Notably, our method is significantly more efficient than previous studies. For instance, [11] required 20 s to obtain a single pair of deformed data, [16] needed 62.32 s to scan a 3 cm trajectory on a flat surface, and [12] required at least 25 s to execute and collect a cycle of palpation data. In contrast, our method enables wide-area robotic screening using QE with improved efficiency.

We assessed the force control performance across both 'static' and 'screening' motion patterns. For 'static' imaging evaluation, we randomly selected ten different positions on phantoms with various elasticity and curvature, performing twenty palpation cycles at each. The axial variations of the desired and applied palpation forces on the abdominal and thyroid phantoms are shown in Fig. 7(b) and (c), respectively. The comparison results demonstrate that the proposed method effectively regulates the probe to apply a sinusoidal-like axial force on soft surfaces for palpation. As listed in Table II, in-plane force control exhibits higher precision in 'static' scenarios compared to 'screening', with lower force errors. Ideally, non-axial force components (F_x and F_z) should remain close to zero during palpation to minimize non-axial displacement in tissues. The curved soft surface of the thyroid phantom poses challenges to adaptive adjustment during screening, resulting in less accurate perpendicular posture and a higher axial palpation force error and x-axis force. Compared to existing systems, [13] reported force errors ranging between ± 1 N on a flat phantom, while in [36], the elevational (F_x) and lateral (F_z) force were 0.5 ± 0.1 N and 0.8 ± 0.2 N, respectively, in non-screening scenarios. Our method demonstrates acceptable performance in both force control accuracy and efficiency, effectively supporting the probe operations required for QE.

C. Evaluation of Strain Estimation

We evaluated the strain estimation performance in terms of both displacement estimation and strain quality. Two evaluation metrics commonly used were measured: (1) SSIM [29] to measure structural differences between the warped moving image and the fixed image, and (2) the Jacobian folding coefficient (JAC) [37] to quantify the regularity by calculating the percentage of folded pixels in the Jacobian determinant of the deformation fields (i.e., $|J_\phi| \leq 0$). To quantitatively assess strain quality, we calculate the signal-to-noise ratio (SNR) and the contrast-to-noise ratio (CNR) [38] as evaluation criteria:

$$SNR = \frac{\bar{\varepsilon}_t}{\sigma_t} \quad CNR = \sqrt{\frac{2(\bar{\varepsilon}_b - \bar{\varepsilon}_t)^2}{{\sigma_b}^2 + {\sigma_t}^2}} \tag{22}$$

where $\bar{\varepsilon}_t$ and $\bar{\varepsilon}_b$ are the average strains in the target and background regions, and σ_t and σ_b are the standard deviations of strains in those regions, respectively. SNR and CNR are popular metrics used to assess elastography methods in the absence of ground truth (GT) strain values. SNR measures the overall quality of strain estimation, while CNR indicates the detectability of lesions. The lesion region is defined based on manual segmentation. We calculate the length r of the minor axis of the mask and define the region length as $l = \alpha \times \sqrt{2}r/2$, where $\alpha = 0.8$ to be statistically meaningful, as depicted in the first column of Fig. 8. We also segmented a vessel structure in the abdominal phantom and the trachea in the thyroid phantom for comparison. This metric CNR2 indicates the method's ability to distinguish lesions from normal tissue distractors.

We compared the proposed transformer-based displacement estimation method with the classical LK [32] and state-of-the-art deep learning-based approaches. We trained the representative CNN-based VoxelMorph (VM) [39] and the recurrent network-based RAFT [40] from scratch under the same settings for fair comparison. We carried out experiments with different ρ_s

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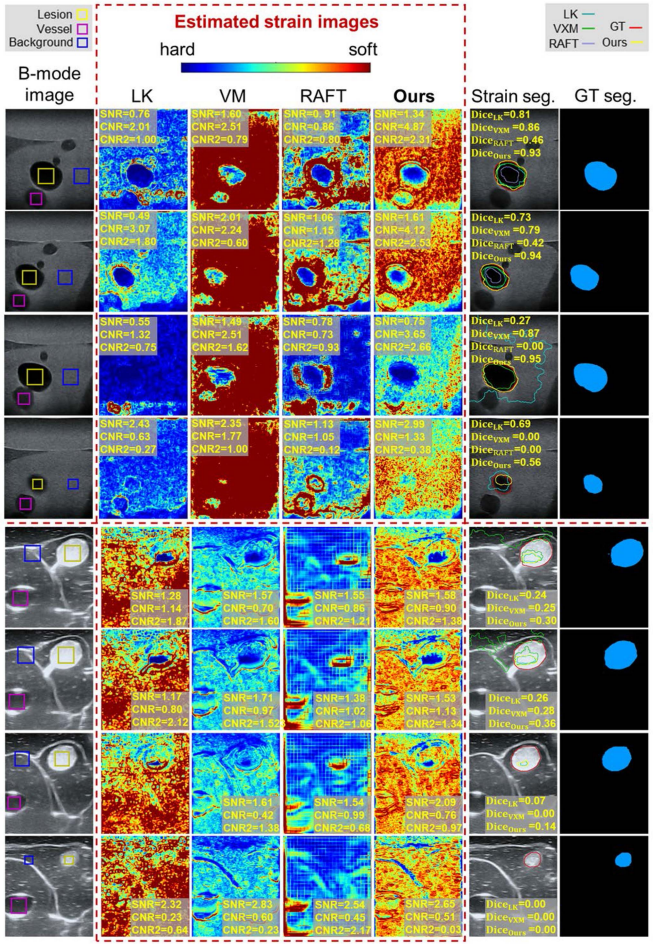


Fig. 8. Visualization and comparison of estimated strain outcomes. The first four rows show results from the abdominal phantoms, while the last four rows show results from the thyroid phantom.

values (5%, 10%, avg., and max. of the maximum SSIM) to validate multi-frame fusion. Quantitative results are listed in Table III, and strain results are normalized and visualized using a colormap in Fig. 8. Our method outperforms both displacement estimation and strain estimation tasks on the abdominal phantom, achieving the highest average displacement estimation accuracy (SSIM $= 0.90 \pm 0.03$), the highest SNR (2.33 ± 1.37) and CNR (2.29 ± 1.27) at $\rho_s = 5\%$, and the highest CNR2 (2.04 ± 1.22) at $\rho_s = 10\%$. Despite the flatter soft surface, a wider palpation force range was applied, inducing larger tissue displacements. In this phantom study, the LK algorithm achieves acceptable average SSIM but shows lower values in SNR and CNR2, indicating poorer strain image quality and reduced ability to distinguish anomalies. VM, despite the lowest SSIM, outperforms LK and RAFT in terms of SNR, CNR, and CNR2. RAFT, while showing better displacement estimation results, performs poorly on all strain metrics. This highlights that displacement estimation performance is not strictly indicative of strain estimation quality, especially when non-axial motions inevitably occur during palpation. Thus, although the displacement field is accurately predicted, out-of-plane components adversely affect strain estimation. Strain images from RAFT show large contrasts between the lesion and adjacent tissues but fail to capture the elasticity relationships with more distant regions.

Degraded performance is observed on the more challenging thyroid phantom. The curved soft surface complicates posture adjustment, leading to larger out-of-plane motion and increased noise in strain estimation. In this scenario, the proposed method yields the highest average SSIM of 0.88 ± 0.05, with the highest SNR (2.26 ± 0.62) and CNR (0.70 ± 0.45) at the avg. setting. These results indicate that our method can predict precise displacement fields from B-mode images for high-quality strain estimation. Displacement estimation accuracy for LK declines on the thyroid phantom, likely due to the increased control errors on the challenging surface. Although both VM and LK demonstrate better CNR2, LK's strain maps exhibit more severe shrinkage in low-strain regions, while VM shows low contrast between the lesion and superficial tissues, resulting in lower CNR values. Strain images estimated by RAFT exhibit low strain contrast and a strong grid effect, failing to accurately reflect the relative elasticity. Incorrect strain estimation and degraded quality may result from non-ideal motions due to force control errors and inaccurate displacement estimation. Nevertheless, multi-frame fusion significantly improves the quality and lesion detectability of the estimated strain maps by eliminating unsatisfactory displacement estimation results. Overall, our transformer-based method shows superior results in strain estimation, exhibiting better robustness to out-of-plane motion errors and large, nonlinear tissue displacements.

We have added an ablation study comparing w/ and w/o SAM-based auxiliary loss. Table IV summarizes the ablation on the two auxiliary losses. The results show no degradation when enabled and consistent, albeit modest, improvements in strain quality on both phantoms. Combining both auxiliaries keeps SSIM/JAC stable and achieves the best CNR_2 (1.60 ± 0.80), indicating complementary benefits. These results support using SAM only as a gated, training-time regularizer.

D. Evaluation of Anomaly Identification

Areas with lower strain values are segmented to evaluate the method's accuracy in anomaly identification. The strain-based segmentation results are shown in the penultimate column of Fig. 8. We used a region growing algorithm [41] for automatic segmentation. The segmentation masks were then post-processed with threshold filtering to retain only large connected areas. Segmentation results with a Dice score below 0.1 are considered failures in anomaly identification, which are reported as the detection rate (Dect.) in Table V.

The proposed method achieves the highest detection rate and segmentation accuracy in both phantom studies. By generating high-contrast strain maps, it facilitates clear differentiation and segmentation of lesion tissue, enabling automatic localization of suspicious anomalies. In the abdominal phantom, LK results show blurred boundaries and low strain contrast, leading to diminished performance. VM suffers from reduced CNR2 and uneven strain distribution that inadequately covers the entire lesion area. This complicates differentiation from similar normal tissues, often resulting in misdiagnoses (column 4 in Fig. 8) and a lower detection rate. Despite reduced strain contrast due

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TABLE III
EVALUATION OF UNSUPERVISED ULTRASOUND IMAGE DISPLACEMENT ESTIMATION ACCURACY AND STRAIN IMAGE QUALITY

Method	ρ_s	Abdominal phantom					Thyroid phantom				
		SSIM ↑	JAC ↓	SNR ↑	CNR ↑	CNR2 ↑	SSIM ↑	JAC ↓	SNR ↑	CNR ↑	CNR2 ↑
-	-	0.69±0.09	-	-	-	-	0.63±0.10	-	-	-	-
LK [32]	5%	0.89±0.04	0.02±0.01	1.20±0.95	1.69±0.95	0.90±0.65	0.82±0.05	0.01±0.01	1.67±0.52	0.67±0.36	1.05±0.54
	10%	0.87±0.05	0.03±0.02	1.26±1.15	1.89±0.90	0.89±0.67	0.80±0.05	0.01±0.01	1.80±0.60	0.69±0.38	1.18±0.46
	avg	0.83±0.05	0.05±0.03	1.22±1.35	1.70±0.76	0.73±0.57	0.75±0.06	0.02±0.02	1.97±0.69	0.55±0.34	**1.58±0.36**
	max	0.92±0.04	0.01±0.01	1.03±0.61	1.29±0.64	0.81±0.51	0.87±0.05	<0.005	1.39±0.34	0.56±0.29	0.76±0.41
VM [39]	5%	0.75±0.02	<0.005	1.85±0.93	1.85±0.87	1.03±0.84	0.87±0.04	<0.005	1.58±0.37	0.47±0.34	1.10±0.58
	10%	0.72±0.03	0.01±0.01	1.94±0.84	2.03±0.99	1.13±0.97	0.84±0.04	<0.005	1.94±0.54	0.57±0.42	1.30±0.52
	avg	0.71±0.03	0.01±0.01	1.93±1.09	1.98±1.07	1.15±0.97	0.80±0.05	<0.005	2.17±0.69	0.68±0.54	1.15±0.48
	max	0.77±0.01	<0.005	1.52±0.81	1.52±0.62	0.81±0.67	0.91±0.04	<0.005	1.34±0.22	0.42±0.29	0.75±0.43
RAFT [40]	5%	0.91±0.03	<0.005	1.16±0.39	0.94±0.45	0.47±0.33	0.84±0.04	<0.005	1.00±0.24	0.41±0.26	0.79±0.48
	10%	0.89±0.03	<0.005	1.64±0.66	1.55±0.66	0.67±0.40	0.81±0.04	<0.005	1.25±0.38	0.52±0.30	0.88±0.52
	avg	0.88±0.03	<0.005	1.91±0.82	1.94±0.82	0.78±0.42	0.78±0.05	<0.005	1.74±0.49	0.64±0.41	1.14±0.56
	max	0.94±0.02	<0.005	0.86±0.17	0.69±0.28	0.35±0.24	0.88±0.05	<0.005	0.88±0.16	0.35±0.22	0.69±0.45
Ours	5%	0.92±0.02	0.01±0.01	**2.33±1.37**	**2.29±1.27**	1.60±0.80	0.92±0.02	<0.005	1.85±0.48	0.48±0.35	0.88±0.40
	10%	0.91±0.02	0.02±0.01	2.29±1.42	2.04±1.22	**2.04±1.22**	0.90±0.03	<0.005	2.17±0.59	0.63±0.44	0.84±0.42
	avg	0.90±0.03	0.02±0.01	2.29±1.49	1.94±1.18	1.31±0.74	0.88±0.04	<0.005	**2.26±0.62**	**0.70±0.45**	0.88±0.43
	max	0.95±0.02	<0.005	1.52±0.69	1.54±0.84	1.20±0.71	0.95±0.02	<0.005	1.30±0.20	0.38±0.25	0.64±0.34

TABLE IV
ABLATION STUDY ON AUXILIARY LOSSES

$\mathcal{L}_{aux}$ $_{-D}$	$\mathcal{L}_{aux}$ $_{-A}$	SSIM↑	JAC↓	SNR↑	CNR↑	CNR2↑
-	-	0.94±0.02	0.02±0.01	3.80±0.89	1.69±0.69	1.01±0.7
√	-	0.90±0.02	0.01±0.01	1.84±1.18	**2.51±1.24**	1.47±0.7
-	√	0.94±0.02	0.02±0.01	**3.84±0.84**	1.01±0.69	0.95±0.7
√	√	0.92±0.02	0.01±0.01	2.33±1.37	2.29±1.27	**1.60±0.8**

TABLE V
EVALUATION OF ANOMALY IDENTIFICATION ACCURACY

Subject	Term	LK	VM	RAFT	**Ours**
Abdominal phantom	Dect.	0.68	0.62	0.26	**0.77**
	Dice	0.76±0.09	0.81±0.08	0.11±0.20	**0.85±0.10**
Thyroid phantom	Dect.	0.59	0.63	\	**0.73**
	Dice	0.12±0.11	0.16±0.12	\	**0.24±0.19**

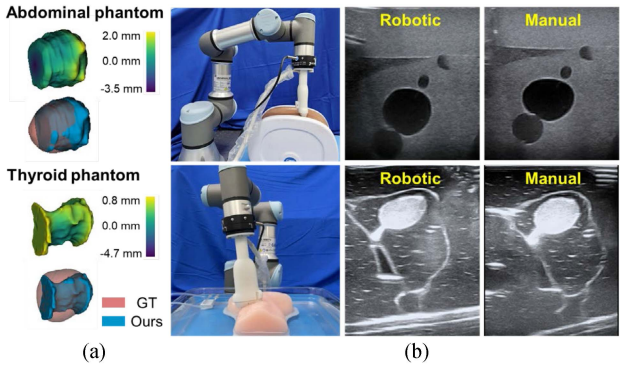


Fig. 9. Results of robotic feedback localization of the anomaly, including the reconstructed 3D anomaly models (a); the robotic localization scene (b); and a sampled comparison of anomaly-centered images (c).

to undesired probe motion (column 3 in Fig. 8), the proposed method still achieves precise anomaly segmentation, while LK and RAFT fail to produce reliable strain results. Although RAFT achieves acceptable displacement estimation performance, its predicted displacement fields are unstable and incapable of producing accurate strain results, failing to identify anomalies on the thyroid phantom.

In the thyroid phantom study, all methods show shrinkage of the estimated lesion region, likely due to the heterogeneity of the lesion material. Strain images from VM exhibit low contrast with the superficial tissue layer, leading to incorrect segmentation and a low Dice score (columns 1, 2 in Fig. 8). Detection failures are observed across all methods at the lesion boundaries (last column in Fig. 8). This limitation can be mitigated in future work by incorporating RF data to enhance sensitivity, especially as these data become more commonly accessible. The above challenges, along with suboptimal in-plane probe motion, lead to an insufficient detection rate and deviations in the statistical strain metrics.

E. Evaluation of Robotic Feedback Localization

After screening, a 3D model of the detected anomaly is reconstructed from the strain-based segmentation masks, as shown in Fig. 9(a). We measured model-to-model distances to assess accuracy. The GT lesion model was reconstructed from B-mode images captured with nearly zero contact force. The robotic probe started from an average initial distance of 48.88 mm for the abdominal phantom and 42.61 mm for the thyroid phantom and was navigated to the predicted lesion target center. Lesion localization accuracy was evaluated by the distance between the lesion's center and the probe's imaging plane centerline. We benchmarked robot control performance against manual operation for comparison. Two medical students were tasked with maneuvering the probe to localize and centralize the lesion in the ultrasound image five times.

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TABLE VI
EVALUATION OF ROBOTIC FEEDBACK LOCALIZATION

Term	Abdominal phantom		Thyroid phantom	
	Robotic	Manual	Robotic	Manual
Model distance [mm]	0.73±	\	1.24±	\
Center error [mm]	1.25±	\	2.05±	\
Localization error	1.01±	**0.84±0.58**	**3.44±**	6.15±0.54
Contact force [N]	4.76±	7.10±0.65	6.14±	3.15±0.66
Time [s]	**3.28±**	27.05±	**7.12±**	22.23±

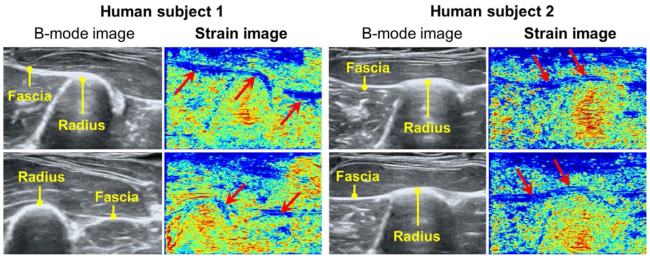


Fig. 10. Robotic QE screening results of the forearms of two human subjects.

As reported in Table VI, our method achieves high accuracy in lesion reconstruction and localization, with average errors of less than 2 mm on the abdominal phantom, meeting requirements for treatment navigation such as guiding needle procedures and ablation [42]. It shows comparable localization precision while requiring eight times less time than manual operation, significantly improving autonomy and efficiency. The localization results show an increased error of 3.44 ± 0.84 mm on the more challenging thyroid phantom. Nevertheless, this level of accuracy can still assist sonographers in quickly localizing suspicious anomalies for further diagnosis and decision-making. As shown in Fig. 9(b), the proposed system enables automatic probe positioning and captures a centered image, similar to manual operation results. Automatic robotic control based on tissue elasticity feedback allows for more precise localization and greater efficiency in more complex situations. Comparatively, the contact force applied using robot control shows a smaller variation across different phantoms. Potential sources of error in robotic anomaly localization include tissue deformation and displacement caused by probe pressure, errors in strain estimation and segmentation, as well as inaccuracies in the feedback control. Overall, our method enables automatic anomaly identification and localization from the robotic scan, distinguishing it from conventional RUISs. *Robotic control sequences on both phantoms and human subjects, are available in the supplementary video.*

F. Demonstration on Human Subjects

We tested the proposed system on human subjects to demonstrate its feasibility in automatically obtaining elasticity information from human tissue. The system was first applied to the forearm of a male and a female volunteer. The displacement estimation model trained on phantom data was directly applied without retraining. The estimated strain images are shown in Fig. 10. Structures with lower elasticity, such as bone and fascia,

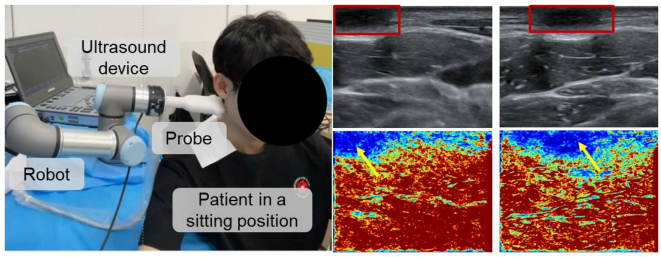


Fig. 11. Robotic QE screening results of the solid subcutaneous mass.

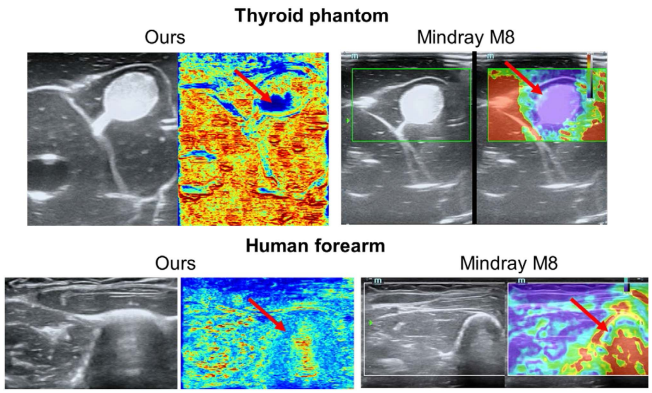


Fig. 12. Comparison of elastography results of a commercial ultrasound system.

are identified in deep blue with low strain values, as indicated by the red arrows. Strain imaging at the bone and soft tissue interface has been investigated to identify bone abnormalities and fractures [43]. Measuring fascia strain has shown poten tial in assisting musculoskeletal diagnosis [44].

We further validated the system's applicability in superficial tissue ultrasound by testing a patient with a solid subcutaneous mass on the neck, as shown in Fig. 11. The lesion, depicted in blue, shows lower elasticity compared to surrounding soft tissue. The proposed method provides visually accurate elasticity information for human tissue. Overall, our system demonstrates effective robotic QE screening on human surfaces, with potential applications in superficial tissue ultrasound examination.

To assess the physical plausibility of our B-mode-based surrogate, we compared it with the strain-elastography output of a commercial RF-based system (Mindray M8, SE option). Fig. 12 shows two representative cases (thyroid phantom and human forearm). Despite the different inputs (display-level B-mode v.s. RF) and effective ROI, both methods delineate stiff regions with lower strain and similar morphology. Our approach provides full-frame strain maps that are further reconstructed into a 3D strain volume for robotic feedback, while the commercial device computes strain only within the ROI. Since the vendor does not expose pixel-level displacement/strain values, we report qualitative agreement.

V. DISCUSSION

In this paper, we introduce a novel robotic ultrasound framework that enables automatic QE screening and anomaly localization. During screening, we propose an efficient force control

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strategy to couple probe motion with tissue elasticity acquisition. It involves an out-of-plane controller for adaptive posture adjustments and an in-plane controller for axial palpation. For localization, we present a Transformer-based unsupervised displacement estimation method to estimate tissue displacements from noisy B-mode images. A strain estimator with multi-frame fusion is introduced to mitigate the negative impacts of control errors for robust strain estimation. Based on the 3D strain map, we implement robotic feedback control for localization and centered imaging of the anomaly.

The proposed method has been quantitatively evaluated on two realistic phantoms and qualitatively tested on human forearms and a subcutaneous mass. Extensive experimental results show that the proposed system can effectively perform QE screening across subjects with varying surface conditions and lesion depths, with improved adaptability and efficiency. It shows promise in pro viding preliminary diagnostic assistance by determining the presence and localization of anomalies based on tissue elasticity. This can help reduce reliance on expert interpretation and enhance accessibility in resource-limited settings.

This study develops a display-level B-mode surrogate for displacement and strain estimation in settings where raw RF is unavailable. Unlike RF-based elastography that exploits phase-sensitive speckle kinematics and is typically evaluated under iso-echoic phantoms, our approach relies on persistent gray-level heterogeneity that remains after beamforming, demodulation, log-compression, and other vendor processing. Consequently, it is not suited to iso-echoic simulation data that intentionally remove intensity contrast. Operating on device output images further implies that the method is applicable only when the lesion is visible on B-mode and is less sensitive to small or visually inconspicuous/iso-echoic lesions. We therefore position the system as an assistive tool for routine superficial examinations, especially for non-expert users, when lesions are visible yet subtle in appearance and morphology relative to surrounding tissue. While ε_{eq} offers direction-agnostic contrast that is beneficial under residual out-of-plane motion, we acknowledge that lateral strain is intrinsically noisier than axial strain and can propagate noise into ε_{eq}. In applications where probe stabilization and tissue coupling tightly constrain out-of-plane motion, axial-only strain remains a practical alternative and may be preferred. Our implementation accommodates either choice by keeping the strain-formation step modular. Looking ahead, we will integrate RF phase information whenever accessible and perform quantitative evaluations on simulation datasets that provide paired RF/envelope and ground-truth displacement/strain. Furthermore, the current system provides only qualitative strain results. Incorporating probe displacement and force measurements or integrating with quantitative methods like SWE could enable more advanced functions such as lesion grading.

Currently, the system has been validated only for superficial lesions at depths of 2∼5 cm, without accounting for large surface motion. The average respiratory period for a resting adult ranges from 3 to 5 s per cycle [45]. Thus, the short palpation cycle (0.95 ± 0.18 s) suggests minimal impact on the in-plane controller. However, the out-of-plane controller is not intended to compensate for large, sudden motion or cyclic respiratory excursions. Therefore, performance may be challenged for deeper abdominal organs due to respiratory motion and force transmission. Future improvements include motion compensation and adaptation to different transducers and moving surfaces. While the average times for displacement estimation (0.040 s), strain estimation (0.007 s), and multi-frame fusion (0.007 s) are relatively short, the total average time required to estimate a strain image is 0.61 s. This duration directly correlates with the image number for multi-frame fusion. Implementing parallel computin g and using more advanced strain estimation methods such as the Savitzky-Golay filter could significantly enhance processing speed. Autonomous screening path planning will also be developed to automate the outer-loop control. Further down the line, its applicability in superficial human tissues such as thyroid nodules, breast lesions, and carotid plaques will be explored through both ex vivo and in vivo patient studies.

VI. Conclusion

We presented a QE-driven robotic ultrasound system that unifies compliant force control with unsupervised, B-mode–only displacement and strain estimation to enable automatic screening and anomaly localization. By maintaining stable probe–tissue contact and generating reliable quasi-static compression, the platform produces robust strain maps and aggregates them into 3D elastic feedback for precise, operator-independent targeting. Experiments on realistic phantoms and proof-of-concept studies on human subjects demonstrate practical deployment, consistent morphology of stiff regions across acquisition conditions, and improved usability compared with conventional grayscale–driven workflows. Centering decisions on tissue elasticity rather than subjective appearance, the system elevates robotic ultrasound from automated scanning to actionable, function-aware imaging, reducing reliance on expert interpretation and laying the groundwork for scalable, routine integration of elastography into autonomous or semi-autonomous clinical workflows.

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